# What does an Algorithmic Choice demonstrate?

A Formal analysis of evidence, Comparison, and preference

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**Abstract**

What does an algorithmic choice demonstrate? An observed output establishes which action a procedure selected, but selection alone does not demonstrate that the action was preferred over a particular alternative. This paper develops a typed evidential framework that distinguishes observed selection, resolved comparison, demonstrated preference, and strategic optimality. A comparison standard specifies admissible evidence certificates, recognized comparative outcomes, and a partial resolution rule. A definitional adequacy lemma establishes that an unresolved comparison precludes demonstrated preference under the declared standard. Finite countermodels delimit unsupported inferences from selection and non-visitation. An analysis of compatible instances establishes a non-identifiability result and an exact local minimax value for two compatible instances under explicit assumptions. An Attribution Audit Record links comparative claims to their evidence sources, certificates, comparison levels, and authorized scopes. Preserved finite-budget decision traces and a separate cross-architecture study motivate the framework, but neither is used as a premise of its formal results.

**Keywords:** formal logic; algorithmic decision-making; computational epistemology; formal methods; evidence-conditional attribution; resolved comparison; decision provenance; identifiability.

## 1 The question

*Does selecting $A$ allow us to conclude that an algorithm prefers $A$ to $B$?*

When an algorithm selects $A$, only one conclusion is immediately certain: $A$ was selected. It does not follow from this fact alone that $A$ was compared with $B$ or judged preferable to it. That stronger conclusion requires evidence that resolves the comparison between the two actions under an explicit standard.

### 1.1 What does the machine's choice allow us to know?

Consider two procedures facing the same alternatives, $A$ and $B$. Both select $A$. Thus far, the observed result is identical. The first procedure selects $A$ only because a fixed rule directs it to choose the first available option. The second selects $A$ after comparing the two alternatives by means of admissible evidence. From the selection of $A$ alone, we may therefore conclude only that $A$ was selected. Only in the second case may we conclude that $A$ was preferred to $B$, because only there does the available evidence establish that comparative claim.

*This article reflects the author's own research and conclusions.

### 1.2 From what is observed to what may be demonstrated

This investigation begins with a fundamental observation: we can observe which action a machine selects, but not necessarily what it has demonstrated a preference for. An observed selection is not, by itself, sufficient to establish that the selected action was compared with a particular alternative. My proposal is to formalize this distinction and to specify the evidence required to justify such an attribution. Let $b$ denote the selected action, $a$ a particular alternative, $h$ the information available to an observer, and $C$ a declared evidential standard. The central semantic non-entailment is

$$\{\mathrm{Selected}_h(b)\} \not\models \mathrm{DP}_h^C(b, a). \tag{1}$$

In ordinary language, observing that a procedure selected $b$ is not, by itself, sufficient to establish that the available evidence demonstrates a preference for $b$ over $a$. Equation (1) does not state that $b$ is inferior, that $a$ is better, or that the decision is incorrect. It establishes a narrower boundary: the fact that an action was selected and the claim that it was demonstrably preferred are different kinds of statements.

### 1.3 From selection to evidential attribution

I began this investigation with a simple question: what, exactly, can be inferred when an algorithm selects one action rather than another? The initial observations drew attention to limited direct exploration. Yet that explanation proved too narrow: different procedures may resolve comparisons through samples, evaluations, admissible bounds, proofs, or other evidence certificates.

Rather than asking, "What did the algorithm prefer?", I therefore ask: "What comparative claim is justified by the available evidence?" The purpose of this paper is not to attribute hidden intentions to algorithms, but to determine which comparative claims are warranted by the evidence recorded for a decision.

### 1.4 Method and organization

The framework represents evidence in an information state through typed certificates and evaluates comparative claims under an explicit comparison standard. It then separates semantic non-entailment from syntactic derivability, constructs countermodels to common overclaims, and studies identifiability under compatible instances.

The paper first positions the problem in the relevant literature, then defines the formal setting and proves the attribution results. Later sections address semantic non-entailments, compatible-instance identifiability, a local minimax consequence, comparison levels, and a concrete audit record. Concise applications illustrate the framework without controlling its general theory.

**Companion study.** A separate controlled study examines how UCT, PUCT, truncated Alpha–Beta, Beam Search, and Greedy Search produce and use evidence that may resolve comparisons. Its empirical findings are not premises of the formal results established here.

## 2 Related work and scientific positioning

### 2.1 Choice, consideration, and revealed preference

Classical revealed-preference theory connects observed choices to preference relations under structural conditions on choice functions and feasible sets [16, 17]. Work on limited and random attention shows more directly that observed choice need not reveal comparison with every

available alternative because the consideration set may be restricted or unobserved [12, 5]. The present framework does not replace those models. It transfers the core evidential caution to algorithmic traces and asks which recorded evidence licenses a particular comparative attribution.

### 2.2 Preference inference and identifiability

Inverse reinforcement learning and related preference-inference problems show that observed behavior can remain compatible with multiple reward functions or planning models [14, 2]. Best-arm identification develops information-theoretic lower bounds and alternative-instance arguments for deciding which action is best under limited information [10, 7]. The compatible-instance result in this paper uses the same broad indistinguishability logic; no claim of a new mathematical technique is made.

### 2.3 Explanation fidelity and provenance

Explainable-AI research distinguishes a report of an output from a faithful account of the mechanism that produced it [13, 6, 8, 9, 19]. Decision provenance similarly motivates recording the data, transformations, and decision pipeline needed for later audit [18]. The contribution here is narrower: a formal test for when comparative preference language is licensed by an explicit evidence standard, together with a minimal audit record for that claim.

### 2.4 Bounded computation and search

Metareasoning studies how computational limits shape the information acquired before a decision [15]. UCT and Monte Carlo Tree Search illustrate why architecture-specific evidence matters: search procedures differ in selection, expansion, simulation, backup, and recommendation [11, 3]. Recent explainable-MCTS work extracts explanatory information from tree structure or validates traversed states and actions against temporal-logic specifications [4, 1]. These approaches reinforce the distinction between the availability of trace data and the evidential strength of the comparative claims drawn from it.

### 2.5 Scientific positioning

The contribution is a typed formal framework for judging the warrant of claims about algorithmic selection. It draws on formal logic, partial information, preference theory, and decision provenance; the same definitions can be applied to multiple computational architectures once a domain-specific evidential standard has been declared. Search engines, generative models, and game-tree algorithms are applications of the framework, not its intended subject matter.

No claim of priority is made for modus tollens, indistinguishability arguments, or the general observation that choice need not fully reveal preference. The intended contribution is the integration of typed evidence certificates, explicit comparison standards, formal attribution limits, and an operational audit record.

## 3 Contributions and scope

The paper makes the following formal and methodological contributions:

- an explicit separation among observed selection, resolved comparison, demonstrated preference, and strategic optimality;

- a typed evidential model with certificate space $\mathrm{Cert}(h)$ and a comparison standard $C$ that specifies certificate admissibility, recognized comparative outcomes, and a partial resolution rule;
- a definitional comparative-adequacy lemma, the induced background attribution theory $\Gamma_C$, and an attribution consequence with complete premises;
- a catalog of semantic non-entailments supported by explicit countermodels;
- a compatible-instance non-identifiability theorem and an exact two-instance local minimax consequence;
- a distinction among operationally resolved, statistically supported, and strategically certified comparisons; and
- an Attribution Audit Record linking each comparative claim to its evidence standard, certificate, level, and authorized scope.

### 3.1 Scope of the results

The framework establishes that selection alone does not entail demonstrated preference and that an unresolved comparison precludes demonstrated preference under the declared standard. It does not establish that selection never expresses preference, that non-visitation always implies unresolved comparison, that operational comparison certifies statistical or strategic superiority, or that formal validity constitutes causal explanation, strategic-error diagnosis, independent replication, or established worldwide novelty.

## 4 Formal setting and evidential standard

Let $s$ be a decision state and let $\mathcal{A}(s)$ be a finite, nonempty set of available actions. Let $h \in \mathcal{H}$ be the information state available to the observer. It may include the returned action, execution trace, scores, samples, priors, bounds, pruning certificates, or other declared evidence. Let $\mathrm{Cert}(h)$ denote the typed set of evidence certificates available in $h$. Let $a, b \in \mathcal{A}(s)$ with $a \neq b$.

### 4.1 Comparison standard

A comparison standard is a tuple

$$C = (\mathrm{Adm}_C, \mathcal{R}_C, \mathrm{Resolve}_C). \tag{2}$$

For $\kappa \in \mathrm{Cert}(h)$, $\mathrm{Adm}_C(\kappa; h, b, a)$ states whether $\kappa$ is admissible for comparing $b$ with $a$. The set $\mathcal{R}_C$ contains the comparative outcomes recognized by the standard. The partial function

$$\mathrm{Resolve}_C : \mathrm{Cert}(h) \times \mathcal{A}(s) \times \mathcal{A}(s) \rightharpoonup \mathcal{R}_C$$

returns a recognized outcome when the certificate resolves the comparison; undefinedness represents an unresolved comparison.

A certificate may comprise samples, a confidence statement, a direct evaluation, an admissible bound, a proof object, an oracle result, or another explicitly declared evidential object. Membership $\kappa \in \mathrm{Cert}(h)$, rather than set inclusion $\kappa \subseteq h$, is used throughout.

**Definition 4.1** (Observed selection)**.** $\mathrm{Selected}_h(b)$ holds exactly when $h$ records $b$ as the action returned or recommended by the procedure. It is an output fact, not a pairwise preference relation.

**Definition 4.2** (Resolved comparison)**.**

$$\mathrm{RC}_h^C(b,a) \iff \exists \kappa \in \mathrm{Cert}(h) : \mathrm{Adm}_C(\kappa; h, b, a) \wedge \mathrm{Resolve}_C(\kappa; b, a)\!\downarrow. \tag{3}$$

Resolution may favor $b$, favor $a$, establish equivalence, or return another recognized outcome.

**Definition 4.3** (Demonstrated preference)**.**

$$\mathrm{DP}_h^C(b,a) \iff \exists \kappa \in \mathrm{Cert}(h) : \mathrm{Adm}_C(\kappa; h, b, a) \wedge \mathrm{Resolve}_C(\kappa; b, a) = b \succ_C a. \tag{4}$$

**Definition 4.4** (Global demonstrated preference)**.**

$$\mathrm{GDP}_h^C(b) \iff \forall a \in \mathcal{A}(s),\ a \neq b \Rightarrow \mathrm{DP}_h^C(b,a).$$

**Definition 4.5** (Unresolved-comparison set)**.**

$$\mathrm{UR}_h^C(b) := \{a \in \mathcal{A}(s) \setminus \{b\} : \neg\, \mathrm{RC}_h^C(b,a)\}.$$

**Definition 4.6** (Demonstrated-preference set)**.**

$$\mathrm{PD}_h^C(b) := \{a \in \mathcal{A}(s) \setminus \{b\} : \mathrm{DP}_h^C(b,a)\}.$$

**Definition 4.7** (Strategic optimality)**.** For a fully specified instance $\mathcal{I}$, let $A^\star(\mathcal{I}) \subseteq \mathcal{A}(s)$ be the nonempty set of actions optimal under an independently declared objective. Strategic optimality is not defined by observed selection or by an operational score.

### 4.2 Auxiliary predicates

Where $h$ contains the corresponding trace fields, $\mathrm{NotVisited}_h(a)$ means that the recorded visit count of $a$ is zero. $\mathrm{Rejected}_h(a)$ means that $h$ contains an explicit rejection certificate under a declared rejection rule. $\mathrm{IncompleteCoverage}(h)$ means that at least one available action is unvisited. For a specified instance $\mathcal{I}$ and selected action $b$, $\mathrm{StrategicCorrect}_{\mathcal{I}}(h)$ means $b \in A^\star(\mathcal{I})$, while $\mathrm{StrategicError}_{\mathcal{I}}(h)$ means $b \notin A^\star(\mathcal{I})$. Let $E_s$ denote any declared immediate or local evaluator used in a countermodel.

## 5 Definitional adequacy and attribution consequences

**Lemma 5.1** (Definitional Comparative Adequacy)**.** *For every distinct $a, b \in \mathcal{A}(s)$,*

$$\mathrm{DP}_h^C(b,a) \Rightarrow \mathrm{RC}_h^C(b,a). \tag{5}$$

*Proof.* If $\mathrm{DP}_h^C(b,a)$ holds, definition 4.3 supplies a certificate $\kappa \in \mathrm{Cert}(h)$ such that

$$\mathrm{Adm}_C(\kappa; h, b, a) \quad \text{and} \quad \mathrm{Resolve}_C(\kappa; b, a) = b \succ_C a.$$

The resolution is therefore defined, and definition 4.2 yields $\mathrm{RC}_h^C(b,a)$. □

Thus, as binary relations on the available actions,

$$\mathrm{DP}_h^C \subseteq \mathrm{RC}_h^C\,. \tag{6}$$

The converse does not follow because a resolved comparison may favor the alternative, establish a tie, or return another recognized outcome.

Let $\Gamma_C$ denote the set of all pairwise instances of lemma 5.1:

$$\Gamma_C := \{\mathrm{DP}_h^C(b,a) \to \mathrm{RC}_h^C(b,a) \mid a, b \in \mathcal{A}(s),\ a \neq b\}.$$

**Proposition 5.2** (Definitional Attribution Consequence)**.**

$$\Gamma_C \cup \{\neg\,\mathrm{RC}_h^C(b,a)\} \vdash \neg\,\mathrm{DP}_h^C(b,a). \tag{7}$$

*Proof.* The theory $\Gamma_C$ contains $\mathrm{DP}_h^C(b,a) \rightarrow \mathrm{RC}_h^C(b,a)$. Together with $\neg\,\mathrm{RC}_h^C(b,a)$, modus tollens yields $\neg\,\mathrm{DP}_h^C(b,a)$. □

*Remark* 5.3 (Evidential boundary)*.* The proposition does not assert that $b$ is inferior, that $a$ is optimal, or that the procedure failed. It establishes only that the available evidence does not demonstrate the stated preference under the declared standard.

### 5.1 Corollaries

**Corollary 5.4** (Selected but not demonstrably preferred)**.**

$$\Gamma_C \cup \{\mathrm{Selected}_h(b), \neg\,\mathrm{RC}_h^C(b,a)\} \vdash \mathrm{Selected}_h(b) \wedge \neg\,\mathrm{DP}_h^C(b,a).$$

**Corollary 5.5** (An unresolved alternative blocks global demonstrated preference)**.**

$$\Gamma_C \cup \{\mathrm{UR}_h^C(b) \neq \varnothing\} \vdash \neg\,\mathrm{GDP}_h^C(b).$$

**Corollary 5.6** (Global demonstrated preference requires complete resolution)**.**

$$\Gamma_C \cup \{\mathrm{GDP}_h^C(b)\} \vdash \mathrm{UR}_h^C(b) = \varnothing.$$

**Corollary 5.7** (Cardinality bound)**.** *Under* $\Gamma_C$*,*

$$\Gamma_C \vdash \left|\mathrm{PD}_h^C(b)\right| \leq |\mathcal{A}(s)| - 1 - \left|\mathrm{UR}_h^C(b)\right|. \tag{8}$$

*Proof.* Under $\Gamma_C$, the sets $\mathrm{PD}_h^C(b)$ and $\mathrm{UR}_h^C(b)$ are disjoint subsets of $\mathcal{A}(s) \setminus \{b\}$. Hence $|\,\mathrm{PD}_h^C(b)| + |\,\mathrm{UR}_h^C(b)| \leq |\mathcal{A}(s)| - 1$, which rearranges to (8). □

Taken together, the corollaries show that selection may coexist with unresolved comparisons; one unresolved alternative blocks global demonstrated preference; global demonstrated preference requires complete resolution; and unresolved comparisons bound the number of demonstrable pairwise preferences.

## 6 Semantic non-entailments

The following propositions concern semantic consequence. Each is established by an explicit countermodel in section B: a model satisfies the premise while falsifying the proposed conclusion.

**Proposition 6.1** (Selection alone is insufficient)**.**

$$\{\mathrm{Selected}_h(b)\} \not\models \mathrm{DP}_h^C(b,a). \tag{9}$$

**Proposition 6.2** (Non-visitation does not entail non-resolution)**.**

$$\{\mathrm{NotVisited}_h(a)\} \not\models \neg\,\mathrm{RC}_h^C(b,a).$$

**Proposition 6.3** (Non-visitation does not entail rejection)**.**

$$\{\mathrm{NotVisited}_h(a)\} \not\models \mathrm{Rejected}_h(a).$$

**Proposition 6.4** (Immediate maximization does not entail strategic optimality)**.**

$$\left\{ a \in \arg\max_{x \in \mathcal{A}(s)} E_s(x) \right\} \not\models a \in A^\star(\mathcal{I}).$$

**Proposition 6.5** (Coverage does not determine strategic correctness)**.**

$$\{\text{IncompleteCoverage}(h)\} \not\models \text{StrategicError}_{\mathcal{I}}(h),$$
$$\{\neg\, \text{IncompleteCoverage}(h)\} \not\models \text{StrategicCorrect}_{\mathcal{I}}(h).$$

These propositions delimit distinct overclaims. Selection does not supply a pairwise certificate; zero visits do not identify the evidential status of a comparison; omission is not rejection; a local maximizer need not be strategically optimal; and coverage alone does not settle correctness.

# 7 Compatible-instance identifiability and a local minimax consequence

Let $\mathfrak{I}(h)$ be the class of fully specified decision instances compatible with the same information state $h$. Define uniform best-action identifiability by

$$\text{UBAI}(h) \iff \bigcap_{\mathcal{I} \in \mathfrak{I}(h)} A^\star(\mathcal{I}) \neq \varnothing. \tag{10}$$

**Theorem 7.1** (Compatible-Instance Non-Identifiability)**.** *If* $\mathcal{I}^-, \mathcal{I}^+ \in \mathfrak{I}(h)$ *and*

$$A^\star(\mathcal{I}^-) \cap A^\star(\mathcal{I}^+) = \varnothing,$$

*then* $\neg\, \text{UBAI}(h)$.

*Proof.* Suppose UBAI($h$). Then some action belongs to $A^\star(\mathcal{I})$ for every $\mathcal{I} \in \mathfrak{I}(h)$. In particular, it belongs to both $A^\star(\mathcal{I}^-)$ and $A^\star(\mathcal{I}^+)$, contradicting the assumed disjointness. □

*Remark* 7.2 (Scope)*.* The theorem concerns identifiability, not search coverage. Incomplete coverage alone neither proves nor disproves UBAI($h$); valid side information may identify an optimum without direct visitation.

Assume now that $\mathcal{I}^-$ and $\mathcal{I}^+$ yield the same information state $h$; the recommendation rule depends only on $h$; the same conditional distribution $q(\cdot \mid h)$ is used in both instances; and the nonempty optimal-action sets are disjoint. Define

$$\text{Err}_j(q \mid h) := \Pr_{\mathcal{I}^j}\Big(\widehat{a} \notin A^\star(\mathcal{I}^j) \mid h\Big), \qquad j \in \{-, +\}.$$

**Theorem 7.3** (Exact Two-Instance Local Minimax Value)**.** *Under the stated assumptions,*

$$\inf_q \max_{j \in \{-,+\}} \text{Err}_j(q \mid h) = \frac{1}{2}. \tag{11}$$

*Proof.* Disjointness gives $q(A^\star(\mathcal{I}^-) \mid h) + q(A^\star(\mathcal{I}^+) \mid h) \leq 1$. Hence at least one success probability is at most $1/2$, so the corresponding error is at least $1/2$. Conversely, choose $x^- \in A^\star(\mathcal{I}^-)$ and $x^+ \in A^\star(\mathcal{I}^+)$, and assign probability $1/2$ to each. Each instance then has success probability $1/2$, so the worst-case error is at most $1/2$. □

For deterministic rules, the local worst-case error is 1. The result is local and conditional; it is not a universal property of search algorithms and is not presented as a new minimax technique.

# 8 Three Standards of warranted Comparison

The framework retains the general predicate $\mathrm{RC}_h^C$ and distinguishes three classes of comparison standards: operational resolution, statistical support, and strategic certification. These distinctions belong to the logic of warranted attribution, regardless of the application architecture.

| **Level and standard** | **Typical certificate** | **Authorized attribution** | **Not automatically authorized** |
|---|---|---|---|
| Operational ($C_{\mathrm{op}}$) | Correct application of a declared internal criterion: local score, visit statistic, prior-guided rule, pruning rule, or tie-break. | The operational criterion favors, disfavors, or ties the alternatives. | Statistical confidence or strategic optimality. |
| Statistical ($C_{\mathrm{stat}}$) | Samples, uncertainty assumptions, and an explicit inferential threshold or decision rule. | The recorded evidence supports a comparative claim at the declared statistical level. | Strategic truth beyond the statistical model and assumptions. |
| Strategic ($C_{\mathrm{strat}}$) | Exact solution, valid oracle, admissible bound, proof certificate, or another objective-valid certificate. | The relevant strategic relation is certified under the declared objective. | Claims outside the certificate's objective or domain. |

The framework assumes no unconditional implication from operational resolution to statistical support, from operational resolution to strategic certification, or from statistical support to strategic certification. Any bridge requires additional domain-specific premises. Local scores, priors, visits, samples, bounds, and strategic proof certificates are not treated as interchangeable.

# 9 Attribution Audit Record and illustrative applications

To evaluate an assertion of the form "the procedure preferred $b$ to $a$", an auditor records the selected output, the compared alternative, the claimed relation, the admissible certificate, the comparison standard, and the precise scope of the resulting attribution.

| # | **Audit field** | **Required content** |
|---|---|---|
| 1 | Selected action | The action returned or recommended. |
| 2 | Comparator action | The alternative against which preference is claimed. |
| 3 | Claimed relation | For example, $b \succ a$, tie, dominance, or strategic superiority. |
| 4 | Comparison standard $C$ | The declared operational, statistical, or strategic standard. |
| 5 | Evidence source | Samples, local evaluation, prior, bound, proof, oracle, or another source. |
| 6 | Resolution certificate $\kappa$ | The trace fragment or certificate satisfying $\mathrm{Adm}_C$ and supporting $\mathrm{Resolve}_C$. |
| 7 | Resolution level | Operational, statistical, or strategic. |
| 8 | Authorized attribution | The strongest comparative statement licensed by the certificate. |
| 9 | Unsupported stronger attribution | A stronger claim that the recorded evidence does not license. |

The following examples are hypothetical. They illustrate the scope of the formal framework and make no claim about the undocumented internal mechanisms of commercial systems.

*Example* 9.1 (Game-tree search: output is not rejection). A finite-budget game-tree procedure returns move $m_1$, while legal move $m_2$ receives no direct root visit. The observation licenses the

claims that $m_1$ was returned and $m_2$ was not directly visited; it does not, by itself, establish rejection:

$$\{\text{Selected}_h(m_1), \text{NotVisited}_h(m_2)\} \not\models \text{Rejected}_h(m_2).$$

A sound bound could even resolve the comparison without directly visiting $m_2$.

*Example* 9.2 (Ranked result: order is not certified quality). Suppose a ranking procedure places document $A$ above document $B$. The output records an ordering and may express the ranking rule's own operational preference. If $C_{\text{quality}}$ is an independently specified quality standard, position alone does not certify superiority under that standard:

$$\{\text{Rank}_h(A) < \text{Rank}_h(B)\} \not\models \text{DP}_h^{C_{\text{quality}}}(A, B).$$

*Example* 9.3 (Generative answer: citation is not comparative certification). Suppose a generative system cites source $A$ but not source $B$. The visible answer does not establish that $B$ was retrieved, considered, and judged inferior:

$$\{\text{Cited}_h(A), \neg\, \text{Cited}_h(B)\} \not\models \text{DP}_h^{C_{\text{source}}}(A, B).$$

Here Cited denotes an observed output feature, not an assertion about inaccessible internal preferences.

*Example* 9.4 (Positive case: warranted attribution). If an available certificate $\kappa$ is admissible under $C$ and the resolution rule establishes that $b$ is preferable to $a$, then the corresponding attribution is demonstrated under that standard:

$$\kappa \in \text{Cert}(h) \wedge \text{Adm}_C(\kappa; h, b, a) \wedge \text{Resolve}_C(\kappa; b, a) = b \succ_C a \Rightarrow \text{DP}_h^C(b, a).$$

The conclusion remains pairwise and standard-relative unless $C$ certifies stronger claims.

## 10 Empirical motivation and companion study

The formal question was motivated by two preserved finite-budget decision traces. One recorded 45 legal root actions and direct visits to 8 of them; the other recorded 26 legal root actions and direct visits to 8. These observations establish only the recorded output and direct root coverage. They do not establish strategic error, the absence of all information about unvisited alternatives, or the absence of a comparison under every admissible evidential standard.

A separate controlled study examines UCT, PUCT, truncated Alpha–Beta, Beam Search, and Greedy Search. The evidence relevant to comparison resolution differs by architecture and declared standard: samples, prior information, local scores, and valid bounds support different kinds of claims. Quantitative results, implementation details, and empirical counterexamples belong to the companion experimental paper rather than to this formal analysis.

Neither the preserved traces nor the companion study is a premise of the lemmas, propositions, corollaries, or theorems established here.

## 11 Reproducibility and formal autonomy

The formal results depend only on the definitions, assumptions, and rules of inference stated in this manuscript. The illustrative applications and empirical motivation are not premises of the proofs. A companion reproducibility paper has been prepared to document the formal artifact associated with this framework, including its verification materials and finite countermodels. The artifact has already been constructed and verified but falls outside the scope of the present article. Its existence should not be interpreted as empirical validation or external generalization of the framework.

## 12 Limitations

- Lemma 5.1 is definitional, and proposition 5.2 is a direct attribution consequence; neither is presented as a novel inference rule.
- The usefulness of $\mathrm{RC}_h^C$ depends on a non-circular, domain-appropriate comparison standard. An overly weak standard can certify unsupported claims; an overly strong standard can reject valid certificates.
- The three comparison levels are categories of standards, not a universal hierarchy. Bridge implications require independent premises.
- Historical visit distributions do not establish causal effects of alternative orderings, larger budgets, or different tie-breaking rules. Such claims require controlled intervention.
- The exact minimax value is restricted to the declared two-instance indistinguishability setting.
- This paper does not claim established worldwide novelty. Related ideas occur in revealed preference, limited attention, inverse preference inference, identification, decision provenance, metareasoning, and explanation fidelity.
- Independent replication, external validation, and empirical prevalence are separate from formal validity.

## 13 Conclusion

*What does an algorithmic choice demonstrate?*

An algorithmic choice establishes which action was selected. But what does that observation tell us about the comparisons performed by the decision procedure, the alternatives considered, and the preference that an observer may legitimately attribute to it?

This paper has examined a problem that arises when algorithmic decisions are interpreted without distinguishing the outcome of a procedure from the comparative evidence supporting that outcome. A selected action may reflect a resolved comparison, a limited search, an ordering convention, or a decision made under incomplete information. The selection itself does not identify which of these circumstances applies.

The central contribution of the paper is therefore not merely the observation that selection and preference are logically distinct. It is the formal characterization of the evidential conditions under which an observed selection can support a justified comparative attribution.

The framework distinguishes three levels: the action selected, the comparative relations established by admissible evidence, and the preference claims that those relations authorize. These levels must not be conflated. In particular, the absence of direct exploration does not necessarily mean that a comparison remains unresolved: a valid bound, proof, or other admissible certificate may resolve it without requiring every alternative to be visited directly.

The formal analysis establishes two controlling results.

**Result 1 — Selection alone does not entail demonstrated preference.**

$$\boxed{\{\mathrm{Selected}_h(b)\} \not\models \mathrm{DP}_h^C(b, a)} \tag{12}$$

*Explanatory decomposition.*

$$\overbrace{\{\mathrm{Selected}_h(b)\}}^{\text{Observed selection}} \quad \overbrace{\not\models}^{\text{Does not entail}} \quad \overbrace{\mathrm{DP}_h^C(b, a)}^{\text{Demonstrated preference}}$$

*Interpretation.* Knowing that action $b$ was selected is not sufficient, by itself, to establish that the available evidence demonstrates a preference for $b$ over alternative $a$ under evidential standard $C$.

This result does not imply that the algorithm failed to compare the alternatives, that its decision was incorrect, or that the selected action was inferior. It establishes that the observed outcome alone does not justify the stronger comparative attribution.

**Result 2 — An unresolved comparison precludes demonstrated preference.**

$$\boxed{\Gamma_C \cup \{\neg\, \mathrm{RC}_h^C(b,a)\} \vdash \neg\, \mathrm{DP}_h^C(b,a)} \tag{13}$$

*Explanatory decomposition.*

$$\overbrace{\Gamma_C}^{\substack{\text{Comparative}\\ \text{adequacy theory}}} \quad \cup \quad \overbrace{\{\neg\, \mathrm{RC}_h^C(b,a)\}}^{\substack{\text{Unresolved}\\ \text{comparison}}}$$

$$\overbrace{\vdash}^{\substack{\text{Syntactic}\\ \text{derivability}}} \quad \overbrace{\neg\, \mathrm{DP}_h^C(b,a)}^{\substack{\text{Preference not}\\ \text{demonstrated}}}$$

*Interpretation.* Under the comparative adequacy theory $\Gamma_C$, if the comparison between $b$ and $a$ remains unresolved under the declared evidential standard, a preference for $b$ over $a$ is not demonstrated under that standard.

This conclusion is deliberately limited. It does not establish that $a$ is preferable to $b$, nor does it establish that the selected action is incorrect. An unresolved comparison identifies a limitation of the available evidence, not an opposite preference.

*Notation.* Here, $h$ denotes the information state, $b$ the selected action, $a$ a particular alternative, and $C$ the declared evidential standard. The symbol $\nvDash$ denotes semantic non-entailment, whereas $\vdash$ denotes syntactic derivability within the stated theory $\Gamma_C$.

These results provide the logical foundation for the paper's broader analysis, but their significance lies in the evidential framework they support.

A comparative attribution requires more than an observed outcome. It requires identifying the relevant alternative, declaring the evidential standard, establishing that an appropriate certificate is available and admissible, and determining whether that certificate resolves the comparison in the direction claimed.

The distinction between selection and comparative resolution is particularly important in procedures operating under limited computational resources. A decision may be produced before all alternatives have been directly explored. Conversely, a comparison may be resolved through admissible evidence without exhaustive enumeration. Search coverage and demonstrated preference are therefore related but distinct questions.

The analysis of identifiability extends this distinction beyond individual comparisons. When the instances compatible with an information state have no common optimal action, that information does not identify an action that is optimal across all compatible instances. The local minimax result makes the limitation quantitative for two indistinguishable instances with disjoint sets of optimal actions, under the stated assumptions and allowing randomized recommendations.

These results do not establish a universal limitation of a particular search algorithm. They establish limitations on what can be identified or guaranteed from a specified information state.

The practical contribution is the Attribution Audit Record, which provides a structured account of the selected action, the alternative under consideration, the claimed relation, the

evidential standard, the supporting certificate, the resolution obtained, and the attribution that the evidence warrants.

Its purpose is not to certify that an algorithm has found the strategically optimal action. It is to make explicit the difference between what the procedure has produced, what the available evidence establishes, and what an observer is entitled to infer.

The framework applies to different computational settings without assuming that their evidence is interchangeable. A search procedure, a ranking system, and a generative model may all produce observable selections, but the mechanisms and certificates capable of supporting comparative claims can differ. A score, a search trace, a bound, and a formal proof do not necessarily justify the same attribution.

The paper's contribution is thus a formal discipline for evaluating comparative claims about algorithmic decisions. Rather than treating the selected outcome as evidence of every preference that might explain it, the framework requires each attribution to be supported by an admissible comparison under a declared standard.

The question is no longer simply what an algorithm selected. It is what the available evidence establishes about that selection, which alternatives have been comparatively resolved, and which conclusions remain beyond the evidence.

**Choice is a fact. Preference is a conclusion.**

*An algorithmic choice tells us what was selected. The available evidence determines which comparative conclusions are justified.*

# References


[1] Ziyan An, Hendrik Baier, Abhishek Dubey, Ayan Mukhopadhyay, and Meiyi Ma. Enabling MCTS explainability for sequential planning through computation tree logic. In *Proceedings of the 27th European Conference on Artificial Intelligence*, pages 4068–4075, 2024. doi:10.3233/FAIA240976; arXiv:2407.10820.

[2] Stuart Armstrong and Sören Mindermann. Occam's razor is insufficient to infer the preferences of irrational agents. In *Advances in Neural Information Processing Systems 31*, pages 5603–5614. Curran Associates, Inc., 2018. arXiv:1712.05812.

[3] Cameron B. Browne, Edward Powley, Daniel Whitehouse, Simon M. Lucas, Peter I. Cowling, Philipp Rohlfshagen, Stephen Tavener, Diego Perez, Spyridon Samothrakis, and Simon Colton. A survey of monte carlo tree search methods. *IEEE Transactions on Computational Intelligence and AI in Games*, 4(1):1–43, 2012. doi:10.1109/TCIAIG.2012.2186810.

[4] Ronit Bustin and Claudia V. Goldman. Structure and reduction of MCTS for explainable-AI. In *Proceedings of the 27th European Conference on Artificial Intelligence*, pages 1246–1253, 2024. doi:10.3233/FAIA240621; arXiv:2408.05488.

[5] Matias D. Cattaneo, Xinwei Ma, Yusufcan Masatlioglu, and Elchin Suleymanov. A random attention model. *Journal of Political Economy*, 128(7):2796–2836, 2020. doi:10.1086/706861.

[6] Finale Doshi-Velez and Been Kim. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*, 2017.

[7] Aurélien Garivier and Emilie Kaufmann. Optimal best arm identification with fixed confidence. In *29th Annual Conference on Learning Theory*, volume 49 of *Proceedings of Machine Learning Research*, pages 998–1027. PMLR, 2016.

[8] Alon Jacovi and Yoav Goldberg. Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4198–4205. Association for Computational Linguistics, 2020. doi:10.18653/v1/2020.acl-main.386.

[9] Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3543–3556. Association for Computational Linguistics, 2019. doi:10.18653/v1/N19-1357.

[10] Emilie Kaufmann, Olivier Cappé, and Aurélien Garivier. On the complexity of best-arm identification in multi-armed bandit models. *Journal of Machine Learning Research*, 17(1):1–42, 2016. arXiv:1407.4443.

[11] Levente Kocsis and Csaba Szepesvári. Bandit based monte-carlo planning. In *Machine Learning: ECML 2006*, volume 4212 of *Lecture Notes in Computer Science*, pages 282–293. Springer, 2006. doi:10.1007/11871842_29.

[12] Yusufcan Masatlioglu, Daisuke Nakajima, and Erkut Y. Ozbay. Revealed attention. *American Economic Review*, 102(5):2183–2205, 2012. doi:10.1257/aer.102.5.2183.

[13] Tim Miller. Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence*, 267:1–38, 2019. doi:10.1016/j.artint.2018.07.007.

[14] Andrew Y. Ng and Stuart J. Russell. Algorithms for inverse reinforcement learning. In *Proceedings of the Seventeenth International Conference on Machine Learning*, pages 663–670. Morgan Kaufmann, 2000.

[15] Stuart Russell and Eric Wefald. Principles of metareasoning. *Artificial Intelligence*, 49(1–3):361–395, 1991. doi:10.1016/0004-3702(91)90015-C.

[16] Paul A. Samuelson. A note on the pure theory of consumer's behaviour. *Economica*, 5(17):61–71, 1938. doi:10.2307/2548836.

[17] Amartya K. Sen. Choice functions and revealed preference. *The Review of Economic Studies*, 38(3):307–317, 1971. doi:10.2307/2296384.

[18] Jatinder Singh, Jennifer Cobbe, and Chris Norval. Decision provenance: Harnessing data flow for accountable systems. *arXiv preprint arXiv:1804.05741*, 2018.

[19] Sarah Wiegreffe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 11–20. Association for Computational Linguistics, 2019. doi:10.18653/v1/D19-1002.

# A Proof dependencies

## A.1 Definitional comparative adequacy and the background theory

By definitions 4.2 and 4.3, every certificate witnessing $\mathrm{DP}_h^C(b,a)$ also witnesses $\mathrm{RC}_h^C(b,a)$ because the outcome $b \succ_C a$ is defined. Hence lemma 5.1 is definitional. The theory $\Gamma_C$ collects all pairwise instances of the lemma. For the derivation in proposition 5.2, the global-theory presentation is equivalent to the local sequent

$$\{\mathrm{DP}_h^C(b,a) \to \mathrm{RC}_h^C(b,a),\ \neg\,\mathrm{RC}_h^C(b,a)\} \vdash \neg\,\mathrm{DP}_h^C(b,a).$$

### A.2 Global demonstrated preference

If $\mathrm{UR}_h^C(b)$ is nonempty, choose $a^\dagger$ in that set. Then $\neg\,\mathrm{RC}_h^C(b, a^\dagger)$; proposition 5.2 gives $\neg\,\mathrm{DP}_h^C(b, a^\dagger)$. Because $\mathrm{GDP}_h^C(b)$ requires $\mathrm{DP}_h^C(b, x)$ for every $x \neq b$, global demonstrated preference fails.

### A.3 Cardinality bound

Under $\Gamma_C$, the sets $\mathrm{PD}_h^C(b)$ and $\mathrm{UR}_h^C(b)$ are disjoint subsets of $\mathcal{A}(s) \setminus \{b\}$. Hence

$$|\,\mathrm{PD}_h^C(b)| + |\,\mathrm{UR}_h^C(b)| \leq |\mathcal{A}(s)| - 1,$$

which rearranges to corollary 5.7.

## B Explicit countermodels

**B1. Selection without demonstrated preference.** A procedure returns the first admissible action $b$. The information state contains no certificate resolving $b$ against $a$. Thus $\mathrm{Selected}_h(b)$ is true, while $\mathrm{RC}_h^C(b, a)$ and $\mathrm{DP}_h^C(b, a)$ are false.

**B2. Unvisited but resolved.** Action $a$ is not expanded, but a valid upper bound proves that it cannot outperform $b$. Thus $\mathrm{NotVisited}_h(a)$ and $\mathrm{RC}_h^C(b, a)$ are both true.

**B3. Unvisited but not rejected.** The budget expires while $a$ remains legal and unprocessed. Thus $\mathrm{NotVisited}_h(a)$ is true and $\mathrm{Rejected}_h(a)$ is false.

**B4. Immediate maximizer but not strategically optimal.** Action $a$ maximizes a local evaluator $E_s$ but leads to lower terminal utility than another action after deeper continuation.

**B5. Incomplete coverage with strategic correctness.** A valid certificate outside direct visitation establishes that the selected action is optimal even though some actions are unvisited.

**B6. Complete coverage with strategic error.** Every root action receives a local score, but the score is strategically misleading; the selected local maximizer is not in $A^\star(\mathcal{I})$.